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Lab 1

problem - Statement

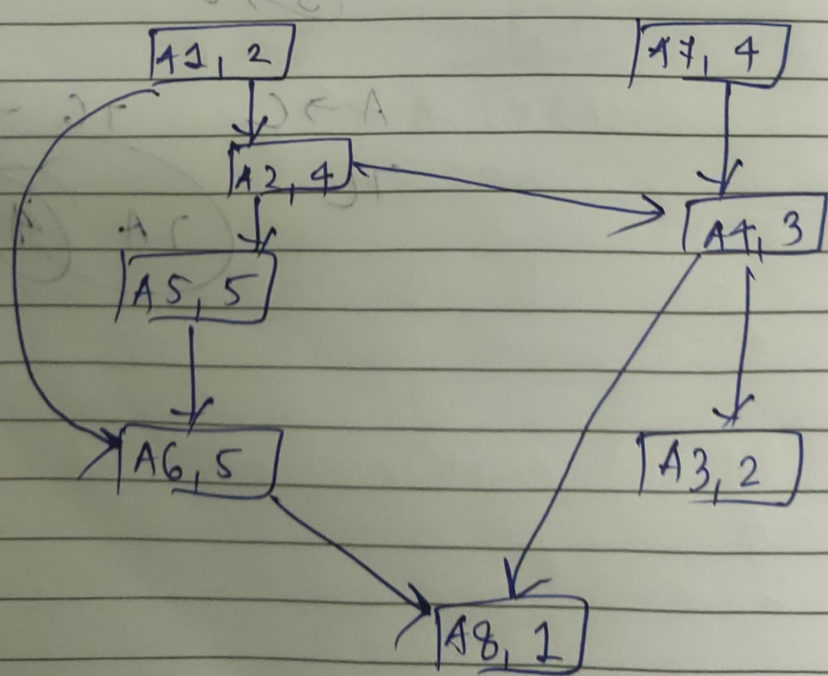
1] Schedule dependent assignments among students with limited daily prompts minimizing days while respecting dependencies and non-partial Completion Constraints.

N → Number of Students in a group

K :- Number of ~~Students~~ prompts per day Each Student

4 1 2 0

A 2 4 1 0



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State:-

State Space (done - assignment , Current - days)

Domain:-

- Assignment = {A1 , A2 . . . An}
- Each assignment has
  - prompt cost
  - Dependency list
- Students = H
- Daily prompt limit per Student = K
- Max days = M

Constraints

1. Dependency Constraint
   - Assignment can start only if all dependencies are Completed.

2. prompt Constraint

   Each student has atmost K prompts/day.

3. No sharing
   - Cannot share prompts.

4. Day limit
   Must Complete within M days.

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Heuristic

1. Reduces Fragmentation.
2. prioritize assignment with long dependency chains.

Time Complexity :- $O(2^n * n^2)$

Code generates all subsets of remaining assignments using bitmasking, validates them with constraints, and recursively builds schedules using DFS with backtracking

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Lab 2

Schedule dependent assignments minimizing Completion time or prompts Considering Student Capacity, dependencies and optional next-day Solution Sharing Constraints.

State-Space

(Completed_assignment, Current_day, mode)

Constraints

1. Dependency Constraint
2. Capacity Constraint
3. Optimization Constraint.

Mode 1 / 3 :- Minimize days (M)

Mode 2 / 4 :- Minimize prompts (K).

Heuristic

1) Reduces Fragmentation.
2) Critical Path Heuristic

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Lab 3

Schedule dependent assignment using two LLm's with cost constraints, minimizing completion time or subscription cost under resource and dependency constraints.

All odd → Gemini C1 → chatgpt cost
All even → chatgpt. C2 → Gemini cost

state-Space = (Completed_assignments, available_assignments, current_day, total_cost, bool LLm's)

Constraints

1. dependency Constraint
2. LLM Constrains.
3. prompt Constraint
   used prompts ≤ daily subscription limits.
4. Case-Based Constraint

Case A:-
|today_set| ≤ numStuden

Case B:-

- Multi assignments allowed
- Sharing only next day.

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Optimization Constraint.

1) Minimize days
2) Minimize cost

Heuristic

A) h(n) = max(remaining_CG_prompt / CG-limit)

remaining_GM_prompt / GM_limit)

remaing_assignment / students

)

B) h(n) = Sum(remaining_prompts × cost_per_prompt)

Approach

1) DFS :- Explore all possibilities.

2) DFBB :- prune if

$g(n) + h(n) \geq$ best-Solution

3) A* Search.

$F(n) = g(n) + h(n)$.

Find optimal Solution Faster.